

Name: Michael Alexaidis
Problem Set Number: #3

Date of Submission: 2/24/2026
Due Date: 2/24/2026

Acknowledgement:
"No help"

1. Known:

- Long, thin rod of radius R (cross-section area $A = \pi R^2$, perimeter $P = 2\pi R$)
- 1D temp variation: $T = T(x, t)$ (radially uniform)
- Convection from lateral surface to ambient T_{∞} , coeff. h

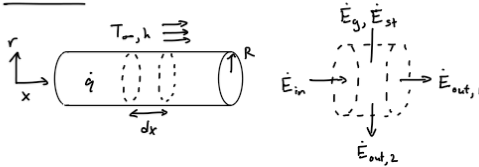
Assumptions:

- Uniform volumetric heat generation $\dot{q}$
- Radiation negligible
- Constant properties: k, ρ, c_p

Find:

- Energy balance on differential control volume dx and express in terms of T
- Governing differential equation for $T(x, t)$
- Boundary and initial conditions for:
 - $x = 0$: maintained at constant temp T_i
 - $x = L$: exposed to air at T_{∞} w/ h
 - $t = 0$: uniform initial temp T_i

Schematic:



Analysis:

(a) General energy balance: $\dot{E}_{in} - \dot{E}_{out} + \dot{E}_g = \dot{E}_{st}$

$$\dot{E}_{in} = \dot{q}_x = -k(\pi R^2) \left. \frac{\partial T}{\partial x} \right|_x \quad (\text{conduction into left face})$$

$$\dot{E}_{out,1} = \dot{q}_{x+dx} = -k(\pi R^2) \left[\frac{\partial T}{\partial x} + \frac{\partial^2 T}{\partial x^2} dx \right] \quad (\text{conduction out of right face})$$

$$\dot{E}_{out,2} = h(2\pi R dx)(T - T_{\infty}) \quad (\text{convection loss from lateral surface})$$

$$\dot{E}_g = \dot{q}(\pi R^2 dx) \quad (\text{heat generation w/in volume})$$

$$\dot{E}_{st} = \rho c_p (\pi R^2 dx) \frac{\partial T}{\partial t} \quad (\text{rate of energy storage})$$

Subbing into energy balance:

$$-k(\pi R^2) \frac{\partial T}{\partial x} + \left[k(\pi R^2) \left(\frac{\partial T}{\partial x} + \frac{\partial^2 T}{\partial x^2} dx \right) \right] - h(2\pi R dx)(T - T_{\infty}) + \dot{q}(\pi R^2 dx) = \rho c_p (\pi R^2 dx) \frac{\partial T}{\partial t}$$

(b) Dividing through by $\pi R^2 dx$:

$$k \frac{\partial^2 T}{\partial x^2} - \frac{2h}{R}(T - T_{\infty}) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$$

(c) $x = 0$: $T(0, t) = T_i, \quad t > 0$

$$x = L: -k \left. \frac{\partial T}{\partial x} \right|_{x=L} = h[T(L, t) - T_{\infty}], \quad t > 0$$

$$t = 0: T(x, 0) = T_i, \quad 0 \leq x \leq L$$

2. Known:

- Wall A: Thickness $L_A = 0.3 \text{ m}$, uniform $\dot{q}$, conductivity k_A
- Wall B: Thickness $L_B = 0.1 \text{ m}$, no $\dot{q}$, conductivity $k_B = 30 \text{ W/mK}$
- Right face of B connects to coolant: $T_{\infty} = 20^\circ\text{C}$, $h = 500 \text{ W/m}^2\text{K}$
- Left surface of A (insulated): $T_w = 115^\circ\text{C}$,
- Interface A-B: $T_i = 100^\circ\text{C}$

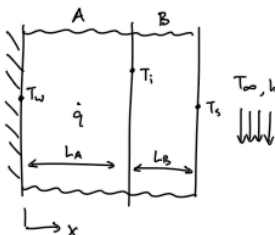
Assumptions:

- Steady State, 1D conduction
- Radiation negligible
- Constant properties
- Perfect thermal contact at interface A-B
- Convection only at right surface of B

Find:

- Right surface temp of wall B: T_s
- Volumetric heat generation in wall A: $\dot{q}$
- Thermal conductivity of wall A: k_A
- Plot $T(x)$ across A and B

Schematic:



Analysis:

(a) In wall B, conduction is linear: $q'' = k_B \frac{T_i - T_s}{L_B}$

At right surface: $q'' = h(T_s - T_{\infty})$

$$k_B \frac{T_i - T_s}{L_B} = h(T_s - T_{\infty}) \rightarrow \frac{30}{0.1} (100 - T_s) = 500(T_s - 20)$$

$$\boxed{T_s = 50^\circ\text{C}}$$

$$q'' = h(T_s - T_{\infty}) = 500(50 - 20) = \boxed{1.5 \times 10^4 \text{ W/m}^2}$$

(b) $q'' = \dot{q} L_A \rightarrow \dot{q} = \frac{q''}{L_A} = \frac{15000}{0.3} = \boxed{5 \times 10^4 \text{ W/m}^3}$

(c) $\frac{dT}{dx} = -\frac{\dot{q}}{k_A}$ BC: $\left. \frac{dT}{dx} \right|_{x=0} = 0, \quad T(0) = T_w$

$$\frac{dT}{dx} = -\frac{\dot{q}}{k_A} x + C_1, \quad C_1 = 0$$

$$T(x) = -\frac{\dot{q}}{2k_A} x^2 + C_2, \quad C_2 = T_w \rightarrow T(x) = T_w - \frac{\dot{q}}{2k_A} x^2$$

Apply $T(L_A) = T_i$:

$$T_i = T_w - \frac{\dot{q}}{2k_A} L_A^2 \rightarrow k_A = \frac{\dot{q} L_A^2}{2(T_w - T_i)}$$

$$k_A = \frac{(50000)(0.3)^2}{2(115 - 100)} = \boxed{150 \text{ W/mK}}$$

(d) See end of document

3. Known:

- Hot gas: $T_{\infty,o} = 1616 \text{ K}$, $h_o = 1000 \text{ W/m}^2\text{K}$
- Cooling air (inside blade): $T_{\infty,i} = 400 \text{ K}$, $h_i = 500 \text{ W/m}^2\text{K}$
- Superalloy (blade): $k_b = 25 \text{ W/mK}$, thickness $L_b = 5 \text{ mm}$
- TBC (ceramic): $k_{TBC} = 1 \text{ W/mK}$, thickness $L_{TBC} = 0.5 \text{ mm}$ (condition 2)
- $R_f'' = 1.0 \times 10^{-4} \text{ m}^2\text{K/W}$ (condition 2)
- $T_{max} = 1200 \text{ K}$

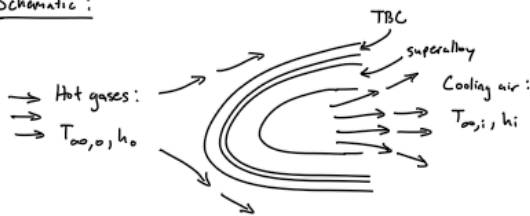
Assumptions:

- Steady state, 1D heat transfer
- Constant properties
- Radiation negligible

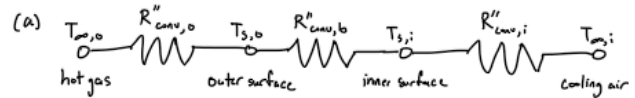
Find:

- (a) No TBC: thermal resistance network, superalloy hot-side surface temp
- (b) With TBC + interface resistance: thermal resistance network, superalloy hot-side surface temp

Schematic:



Analysis:



$$\left. \begin{aligned} R''_{conv,o} &= \frac{1}{h_o} = \frac{1}{1000} = 10^{-3} \text{ m}^2\text{K/W} \\ R''_{conv,b} &= \frac{L_b}{k_b} = \frac{0.005}{25} = 2 \times 10^{-4} \text{ m}^2\text{K/W} \\ R''_{conv,i} &= \frac{1}{h_i} = \frac{1}{500} = 2 \times 10^{-3} \text{ m}^2\text{K/W} \end{aligned} \right\} R''_{total} = 3.2 \times 10^{-3} \text{ m}^2\text{K/W}$$

$$q''_{total} = \frac{T_{\infty,o} - T_{\infty,i}}{R''_{total}} = \frac{1616 - 400}{3.2 \times 10^{-3}} = 3.8 \times 10^5 \text{ W/m}^2$$

$$T_{s,o} = T_{\infty,o} - q'' R''_{conv,o} = 1616 - (3.8 \times 10^5)(0.001) = \boxed{1236 \text{ K}}$$

$$T_{s,o} > T_{max} \text{ so NOT Acceptable}$$



$$\left. \begin{aligned} R''_{conv,o} &= \frac{1}{h_o} = \frac{1}{1000} = 10^{-3} \text{ m}^2\text{K/W} \\ R''_{TBC} &= \frac{L_{TBC}}{k_{TBC}} = \frac{0.0005}{1} = 5 \times 10^{-4} \text{ m}^2\text{K/W} \\ R''_{contact} &= 10^{-4} \text{ m}^2\text{K/W} \\ R''_{conv,b} &= \frac{L_b}{k_b} = \frac{0.005}{25} = 2 \times 10^{-4} \text{ m}^2\text{K/W} \\ R''_{conv,i} &= \frac{1}{h_i} = \frac{1}{500} = 2 \times 10^{-3} \text{ m}^2\text{K/W} \end{aligned} \right\} R''_{total} = 3.8 \times 10^{-3} \text{ m}^2\text{K/W}$$

$$q''_{total} = \frac{T_{\infty,o} - T_{\infty,i}}{R''_{total}} = \frac{1616 - 400}{3.8 \times 10^{-3}} = 3.2 \times 10^5 \text{ W/m}^2$$

$$T_{s,o} = T_{\infty,o} - q''(R''_{conv,o} + R''_{TBC} + R''_{contact})$$

$$T_{s,o} = 1616 - (3.2 \times 10^5)(10^{-3} + 5 \times 10^{-4} + 10^{-4}) = \boxed{1104 \text{ K}}$$

$$T_{s,o} < T_{max} \text{ so Acceptable}$$

